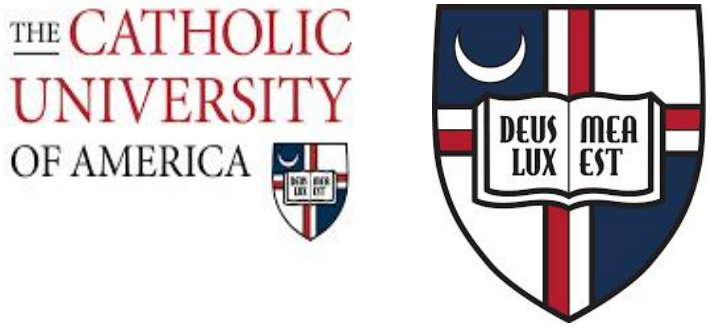
DA - 514

Brain Tumor MRI Classification Using Classical Machine Learning Models



Student: Sandeep Kaur &

Rutwik Devmurari

Instructor: Chaofan Sun

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Abstract

Brain tumors are serious conditions that require timely diagnosis and treatment. Magnetic resonance imaging (MRI) is a standard tool for examining the brain, but interpreting these images manually is labor-intensive and subject to human error. This thesis explores whether several traditional machine-learning algorithms can accurately classify brain MRI images into four categories: *glioma tumor*, *meningioma tumor*, *pituitary tumor*, and *no tumor*. We used a dataset of 7 023 grayscale MRI images (5 712 for training and 1 311 for testing) drawn from public sources. After resizing, normalizing, and flattening the images, we trained and evaluated five models: Logistic Regression, Decision Tree, K-Nearest Neighbors (KNN), Multi-Layer Perceptron (MLP) neural network, and Support Vector Machine (SVM) with a radial basis function kernel. The SVM achieved the highest accuracy of 96.8%, followed by the MLP (95.3%) and Logistic Regression (94.3%). Our findings show that classical models can achieve high performance on brain tumor classification, providing practical alternatives to deep neural networks when data or resources are limited. This work offers a clear pipeline and comparison framework for future studies.

Introduction

Background

Brain tumors can be life-threatening and often require rapid detection to improve patient outcomes. Common types of tumors include gliomas, meningiomas and pituitary adenomas. Gliomas arise from glial cells and are usually malignant, whereas meningiomas originate from the meninges (the protective membranes covering the brain) and are often benign. Pituitary tumors grow in the pituitary gland and can disrupt hormone production. The *no tumor* category refers to healthy brain tissue and serves as a control class.

Magnetic resonance imaging is widely used to visualize brain structures and detect abnormalities. MRIs provide detailed anatomical information, but reading these images manually demands significant time and expertise. In busy clinical settings, automated or semi-automated systems could support radiologists by highlighting potential tumor regions or providing second opinions. Recent advances in artificial intelligence, particularly deep learning, have yielded impressive results in medical image analysis. However, deep neural networks typically require very large datasets and substantial computational resources. This study aims to determine whether simpler models, when applied to a reasonably sized dataset with proper preprocessing, can yield competitive results.

Goals of the Study

The primary goal of this thesis is to develop a complete classification pipeline using classical machine-learning models. Specifically, we aim to:

1. Describe and analyze a multi-class brain MRI dataset containing images of glioma, meningioma, pituitary tumors and normal brains.
2. Outline a simple preprocessing strategy to standardize the images for model training.
3. Train and evaluate five different machine-learning algorithms under the same conditions.
4. Compare their performance using metrics such as accuracy, precision, recall, F1-score, ROC–AUC and log loss.
5. Discuss strengths and weaknesses of each model and suggest directions for improvement.

By completing these steps, we hope to show that classical models remain valuable tools for medical image classification tasks, offering efficiency and interpretability.

Materials and Methods

Dataset

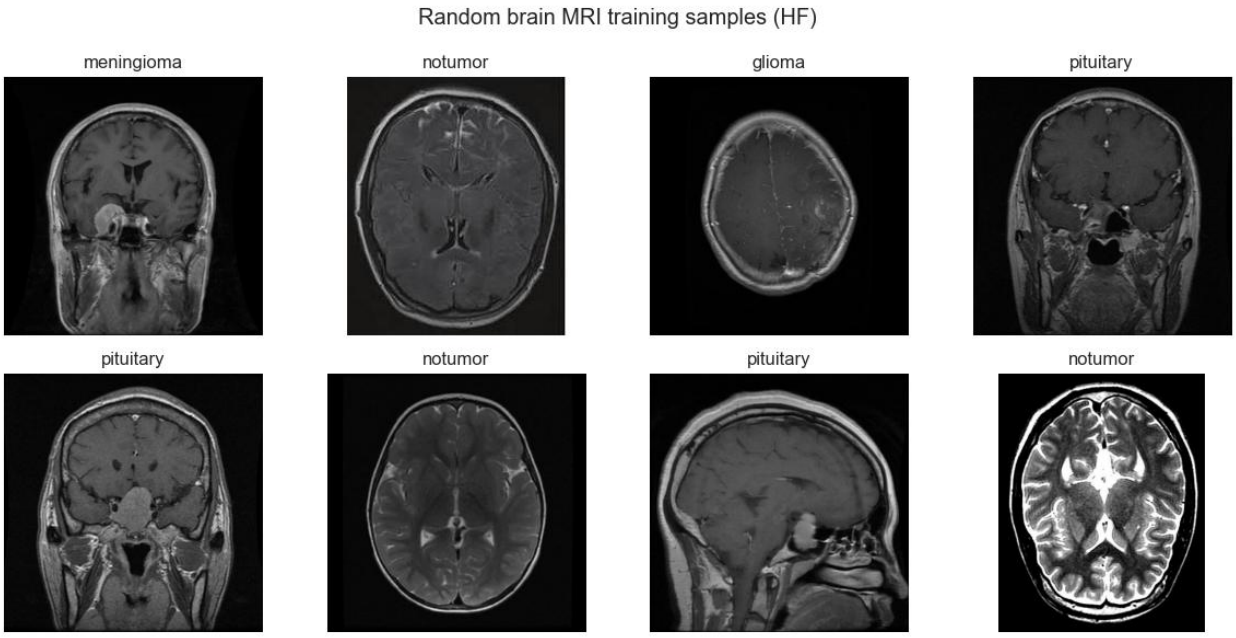
The dataset used in this study consists of 7 023 grayscale axial MRI slices. The images were collected from two public repositories:

(1) A brain tumor dataset published on figshare and

(2) A Kaggle dataset featuring a *no tumor* class.

The combined dataset includes four categories:

- **Glioma** — tumors derived from glial cells.
- **Meningioma** — tumors originating from the meninges.
- **Pituitary** — tumors of the pituitary gland.
- **No tumor** — healthy brain images.



We divided the data into a training set with 5 712 images and a test set with 1 311 images. Table 1 shows the distribution across classes.

Table 1. Class distribution of the training and test sets.

Class	Training images	Test images
Glioma	1321	300
Meningioma	1339	306
Pituitary	1457	300
No tumor	1595	405
Total	**5712**	**1 311**

The roughly balanced class distribution reduces the risk of bias toward any one class during training.

Preprocessing

Before training, all images underwent the following steps:

1. **Re-sizing:** Each image was resized to 128×128 pixels to ensure consistent input size.
2. **Grayscale Verification:** Images were converted to grayscale if they were originally in color, resulting in a single intensity value per pixel.
3. **Normalization:** Pixel values were scaled to the range [0, 1] by dividing by 255. This prevents certain models from being influenced by large numeric ranges and helps with gradient-based learning.

4. **Flattening:** Each 128×128 image was flattened into a one-dimensional vector of length 16 384. This conversion simplifies the input format but discards spatial relationships within the image. Since classical models used here do not explicitly handle spatial structure, flattening is acceptable. Future work could explore handcrafted features like histogram of oriented gradients (HOG), local binary patterns (LBP) or gray-level co-occurrence matrices (GLCM) to capture texture and edge information.

Algorithms Evaluated

We trained and compared the following machine-learning models:

1. **Logistic Regression:** A linear classifier that outputs class probabilities based on the logistic function. Although simple, it often serves as a strong baseline for high-dimensional data.
2. **Decision Tree:** A model that splits the feature space into regions using a series of decisions. Each internal node represents a test on a feature, and leaves correspond to class labels. Decision trees are easy to interpret but can overfit if not constrained.
3. **K-Nearest Neighbors (KNN):** A lazy learning algorithm that assigns the most common label among the k nearest training samples to a new sample. We used $k = 5$ and the Euclidean distance metric.
4. **Multi-Layer Perceptron (MLP):** A neural network with one hidden layer of 100 neurons and ReLU activation. We trained the network with the Adam optimizer for a fixed number of epochs. The MLP provides a non-linear model capable of capturing more complex relationships than logistic regression.
5. **Support Vector Machine (SVM):** A classifier that seeks to maximize the margin between classes. We used the radial basis function (RBF) kernel to enable non-linear decision boundaries. SVMs are known for good performance on high-dimensional data.

Training and Testing

All models were trained on the same training set. We chose default hyperparameters for each algorithm (except for the MLP hidden layer size and KNN neighbor count) to provide a fair comparison. After training, we evaluated each model on the test set using several metrics:

- **Accuracy:** The percentage of test images correctly classified.
- **Precision:** The proportion of correct positive predictions among all positive predictions (macro-averaged across classes).
- **Recall:** The proportion of actual positives correctly identified (macro-averaged).
- **F1-score:** The harmonic mean of precision and recall (macro-averaged).

- **ROC–AUC:** The area under the receiver operating characteristic curve (macro-averaged using a one-versus-rest approach). Higher values indicate better separation between classes.
- **Log Loss:** The cross-entropy loss for the predicted class probabilities. Lower values indicate that a model is making confident and accurate predictions.

These metrics together provide a comprehensive picture of each model's strengths and weaknesses.

Results

Performance Comparison

Table 2 summarizes the performance of each model on the test set. We report accuracy, macro-averaged precision, recall, F1-score, ROC–AUC and log loss.

Table 2. Test performance metrics for the five models.

Model	Accuracy	Precision (macro)	Recall (macro)	F1-score (macro)	ROC–AUC (macro)	Log Loss
Support Vector Machine (SVM)	**0.968**	**0.968**	**0.965**	**0.966**	**0.998**	**0.081**
Multi-Layer Perceptron (MLP)	0.953	0.950	0.949	0.949	0.993	0.195
Logistic Regression	0.943	0.939	0.939	0.939	0.992	0.335
KNN (k = 5)	0.899	0.899	0.894	0.892	0.990	0.343
Decision Tree	0.872	0.868	0.864	0.865	0.911	4.619

The SVM outperformed all other models, achieving the highest values across nearly all metrics. The MLP came in a close second, while logistic regression also performed well considering its simplicity. KNN achieved moderate performance, and the decision tree lagged behind due to its tendency to overfit and produce poorly calibrated probabilities.

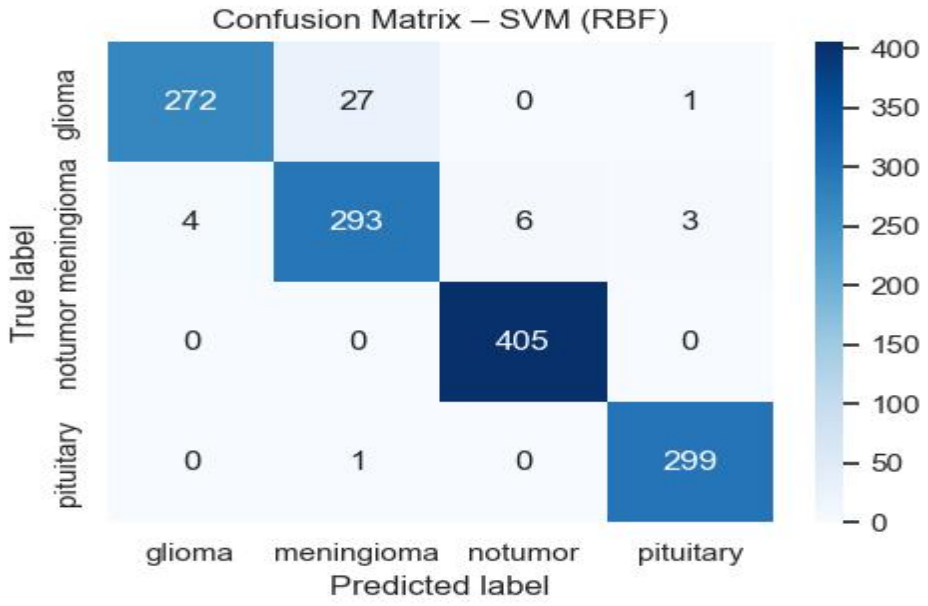
Class Specific Observations

An examination of precision and recall for each class reveals that all models handled the *no tumor* class exceptionally well. This is likely because healthy MRIs lack irregular masses and thus differ markedly from tumor images. The pituitary class was also correctly classified by most models, perhaps because pituitary tumors are distinct in size and location. The more challenging distinction was between glioma and meningioma images. Both tumor types can appear similar in shape and location, causing models to confuse them. The decision tree and KNN, which rely on simple splits and distance measures respectively, made more mistakes in this area. Logistic regression performed surprisingly well, indicating that the data might be almost linearly separable when properly normalized. The MLP and SVM provided the most balanced performance across classes, capturing the subtle differences that separate gliomas from meningiomas.

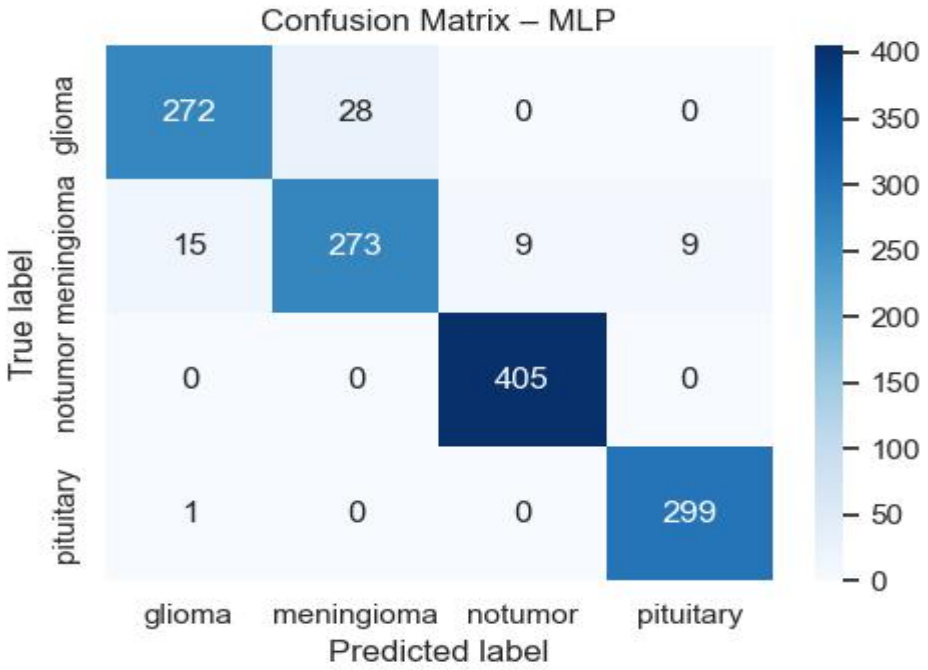
Confusion Matrix Insights

A confusion matrix is a 4×4 table that shows how often a model predicts each class relative to the true class. The diagonal elements represent correct predictions, while off-diagonal elements represent errors. We will see the top two models confusion Matrix here ;

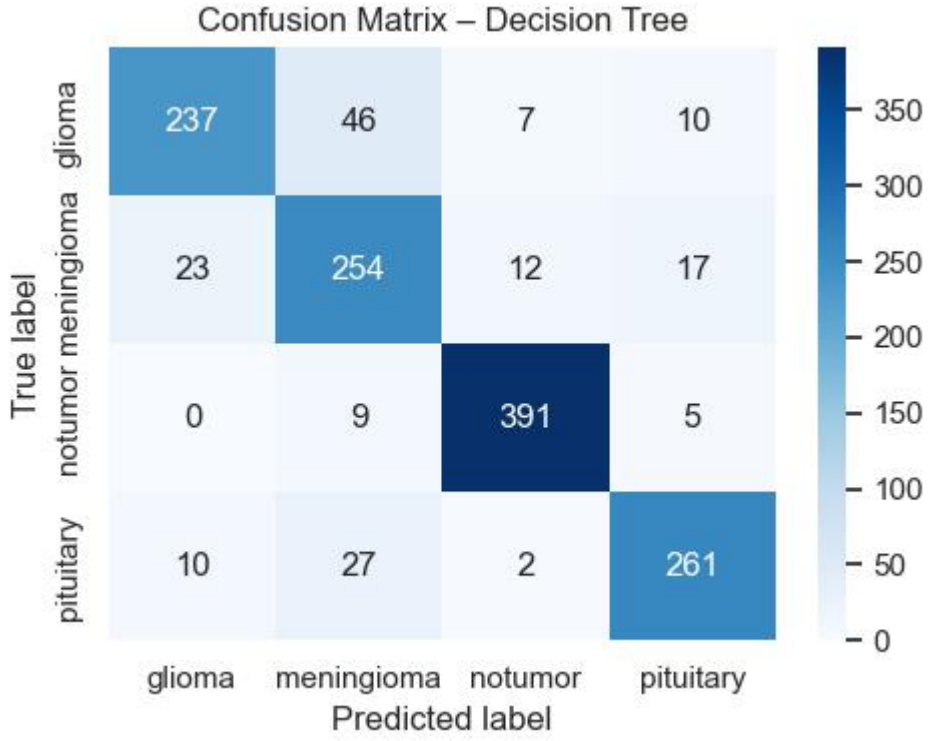
- **SVM:** Nearly all predictions fall on the diagonal. A few meningioma images are misclassified as glioma or vice versa, but errors are rare. No images are misclassified as pituitary or no-tumor.



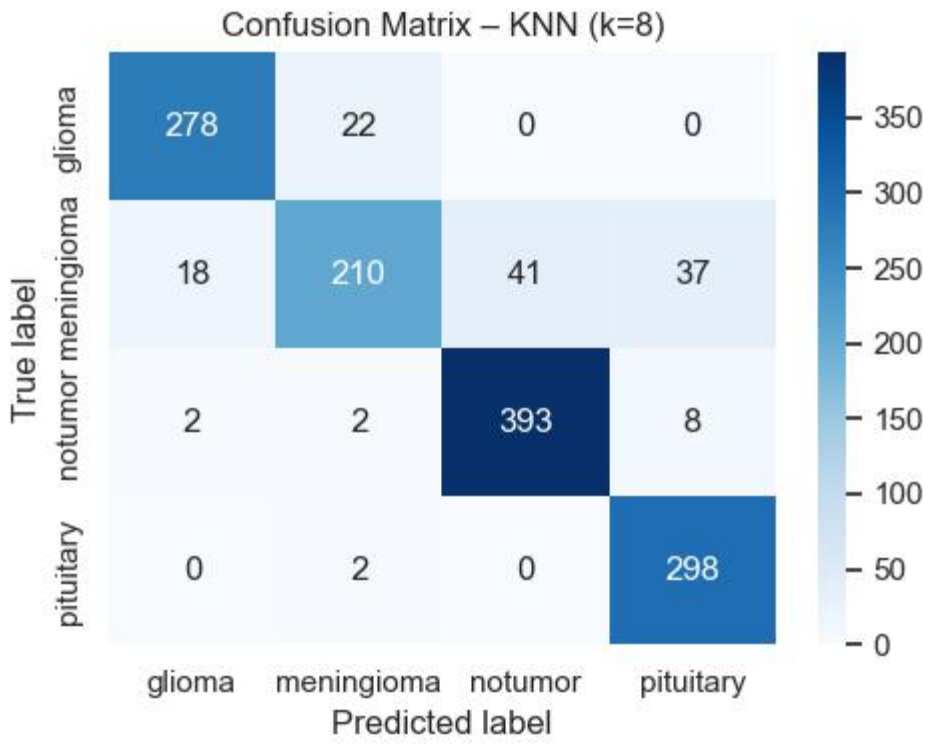
- **MLP:** Similar to the SVM, though with slightly more misclassifications between glioma and meningioma.

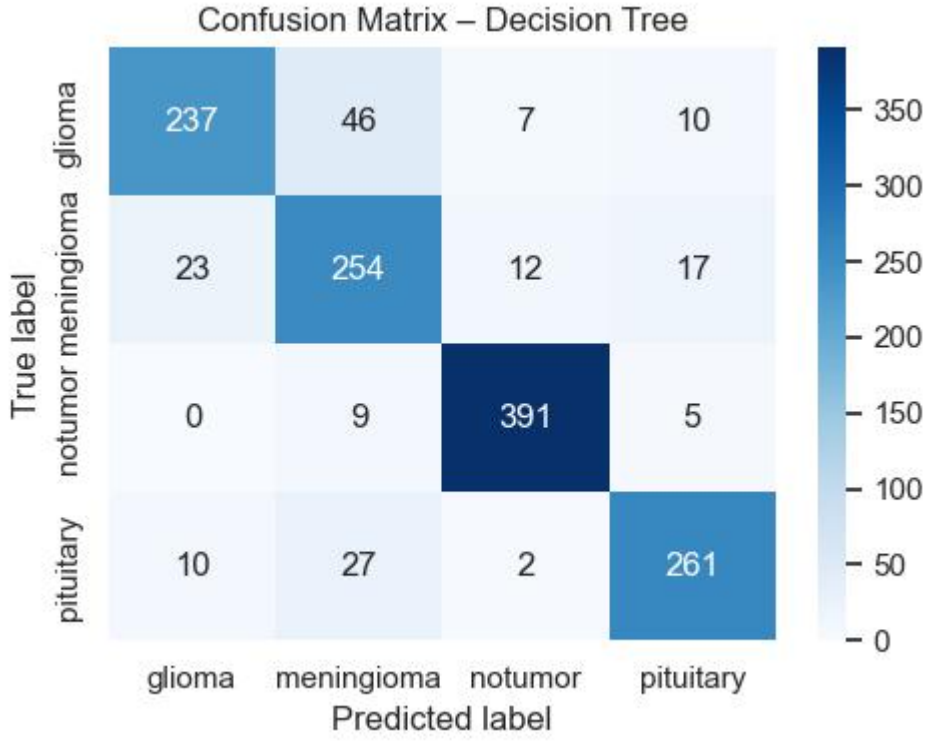


- **Logistic Regression:** More confusion between glioma and meningioma; however, it still rarely mislabels images as pituitary or no-tumor.



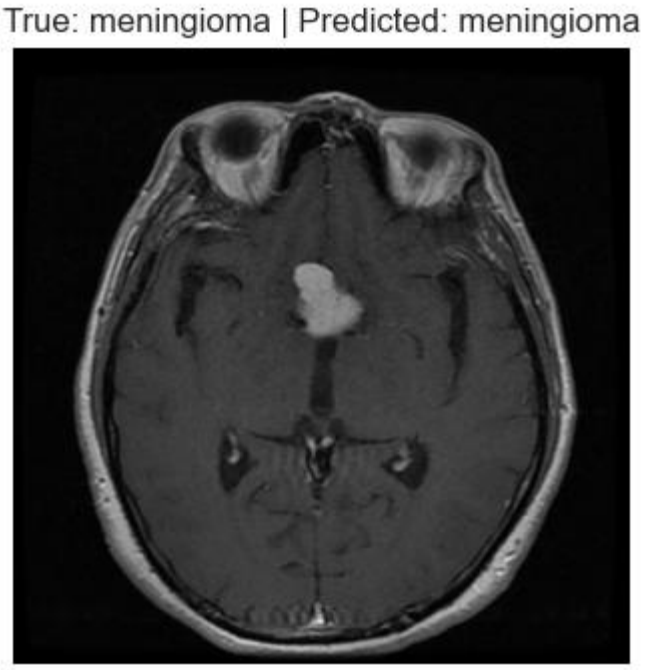
- **KNN and Decision Tree:** These models show more widespread off-diagonal entries. The decision tree, in particular, misclassifies a larger number of images across all classes.





These observations reinforce the idea that non-linear models with a margin-maximizing or hidden layer structure better capture complex patterns in the data.

The Model gives ouptut as it shown in the figure.



Discussion

Implications of Findings

The results demonstrate that classical machine-learning models, particularly the SVM and MLP, can achieve high accuracy on brain tumor classification tasks even when using raw pixel intensities as features. This is encouraging for practitioners in clinical settings who may not have the infrastructure to train deep neural networks. The SVM's ability to handle high-dimensional data and find optimal decision boundaries gives it an advantage in distinguishing subtle differences among tumors. The MLP also performed strongly, benefiting from its non-linear hidden layer. Logistic regression's respectable performance indicates that the four classes are largely separable in the feature space after normalization.

These findings suggest that when computational resources or data are limited, classical models remain relevant. They require less parameter tuning, consume fewer resources and can be trained quickly. They are also more interpretable than deep neural networks—logistic regression coefficients and decision tree splits can provide insight into which pixels contribute most strongly to classification.

Limitations

Despite the promising results, there are limitations. Flattening images discards spatial relationships, which may contain important structural cues. Using more advanced features

or applying techniques like HOG, LBP and GLCM could capture texture and edge information that is lost when flattening. Additionally, we used default hyperparameters; performing cross-validation and hyperparameter optimization could yield further improvements, especially for KNN and decision tree models. Another limitation is that our dataset, while moderate in size, may still be small compared to what is typically used for training deep learning models. More data could improve model generalization and enable deeper networks to be applied effectively.

Furthermore, MRI slices were treated independently, but tumors are three-dimensional structures. Future work could consider using multiple slices or 3-D volumes to capture contextual information along the axial axis. Ensemble methods that combine classical models or integrate deep learning with classical approaches could also provide a balance between performance and interpretability.

Future Directions

To build on this work, future studies could explore:

1. **Feature extraction techniques.** Evaluate HOG, LBP, GLCM, or wavelet features to capture texture and shape information beyond raw intensities.
2. **Ensemble learning.** Combine multiple classifiers (e.g., random forests, gradient boosting) to reduce variance and improve robustness.
3. **Hybrid approaches.** Use deep learning to learn features from MRI slices, then feed these features into classical models for classification. This could yield high performance while maintaining some interpretability.
4. **3-D modeling.** Investigate 3-D CNNs or recurrent neural networks to capture information across consecutive MRI slices.
5. **Clinical integration.** Consider implementing a user interface to help radiologists interact with the model's output, highlight regions of interest and provide confidence scores.

Conclusion

This thesis presented a comparative study of five classical machine-learning models applied to brain MRI classification. Using a dataset of 7 023 images representing four categories (glioma, meningioma, pituitary, and no tumor), we trained and evaluated Logistic Regression, Decision Tree, KNN, MLP and SVM classifiers. The SVM achieved the best test accuracy at 96.8%, followed by the MLP at 95.3%. Logistic regression also performed well, whereas KNN and decision tree were less accurate. Our study demonstrates that carefully prepared classical models can perform remarkably well on brain tumor classification tasks, offering accessible alternatives to more complex deep learning models. Nonetheless, there is room for improvement through feature engineering, hyperparameter tuning and more sophisticated methods. We hope this work serves as a

baseline and encouragement for researchers and clinicians exploring machine learning approaches to medical image analysis.

References

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